1 July 28, 2025 (Class 1)

1.1 Initial Value problem (IVP)

Definition 1.1 (Initial value problem): *An initial value problem (IVP) is a a differential equation together with a specified value, called the initial condition, at a given point in the domain. An IVP is typically expressed in the form:*

$$\begin{aligned} &y'(t) = f(t, y(t)), \quad \text{with } f : \Omega \subseteq \mathbb{R} \times \mathbb{R}^n \to \mathbb{R}^n, \\ &y(t_0) = y_0, \text{where } (t_0, y_0) \in \Omega. \end{aligned}$$

1.2 Hadamard well-posedness of an IVP

For an IVP to be well-posed in the sense of Hadamard, it must satisfy three conditions:

1. The solution exists.
2. The solution is unique.
3. The solution is stable with respect to the data, meaning small changes in the initial conditions or parameters lead to small changes in the solution.

If any of these conditions are not satisfied, the problem is considered ill-posed.

1.3 Lipschitz continuity

Definition 1.2 (Lipschitz continuity.): *Given metric spaces (X, d_X) and (Y, d_Y), a function $f : X \to Y$ is Lipschitz continuous if there exists a constant $L > 0$ such that*

$$d_Y(f(x_1), f(x_2)) \leq L d_X(x_1, x_2), \quad \forall x_1, x_2 \in X.$$

Definition 1.3 (Locally Lipschitz continuity.): *A function $f : X \to Y$ is locally Lipschitz continuous if for every point $x_0 \in X$, there exists a neighborhood U of x_0 such that f is Lipschitz continuous on U, i.e.*

$$\exists\, L > 0, \text{ such that } d_Y(f(x_1), f(x_2)) \leq L d_X(x_1, x_2), \quad \forall x_1, x_2 \in U.$$

1.4 Bounded derivative method of checking Lipschitz continuity

If a function $f : \mathbb{R}^n \times \mathbb{R}^m \to \mathbb{R}^n$ is differentiable on a convex domain $\mathcal{D}$, then it is Lipschitz continuous if,

$$\exists\, L \in \mathbb{R} > 0 \text{ such that } \sup_{(x,y)\in\mathcal{D}} \|\mathcal{J}_f(x, y)\| = L < \infty,$$

where $\mathcal{J}_f(x, y)$ is the Jacobian matrix of f at (x, y) and $\|\cdot\|$ denotes the operator norm.

To note, if the function f is not differentiable, we can still use the definition of Lipschitz continuity to check if it is Lipschitz continuous. If f is not locally Lipschitz on the domain $\mathcal{D}$ of the IVP, then the IVP may not have a unique solution or may not be well-posed.

2 July 30, 2025 (Class 2)

2.1 Picard's theorem (Existence and uniqueness of solutions to IVP)

Let $f : \mathcal{D} \subseteq \mathbb{R}^2 \to \mathbb{R}$ be a real valued f^n of two variables x and y. Let $\mathcal{D}$ be a closed rectangle defined by

$$\mathcal{D} := \{(x, y) \in \mathbb{R}^2 : 0 \le x \le a, 0 \le y \le b, a > 0, b > 0\}.$$

If $f(x, y)$ satisfies the following conditions:

1. $f(x, y)$ is continuous on $\mathcal{D}$ and it is bounded such that $|f(x, y)| \le M$ where $M \in \mathbb{R}$.
2. $f(x, y)$ is locally Lipschitz continuous in y on $\mathcal{D}$, i.e., for every $(x, y) \in \mathcal{D}$, there exists a constant $M > 0$ such that

$$|f(x, y_1) - f(x, y_2)| \le M|y_1 - y_2|, \quad \forall (x, y_1) \in \mathcal{D} \text{ and } (x, y_2) \in \mathcal{D}$$

then the initial value problem (IVP) has a unique solution existing in the interval

$$I : \{x \in \mathbb{R} : |x - x_o| < h\}$$

where $h = \min(a, b/M)$, $M = \max |f(x, y)|$

2.2 Gronwall's lemma

2.3 Approximation methods

2.3.1 Picard's method of successive iteration

Given

$$y' = f(x, y), y(x_o) = y_o, \quad x \in [x_o, a]$$

Let y_o be the initial approximation. Then

$$y_1(x) = y(x_o) + \int_{x_o}^{x} f(x, y_o) dx \quad \text{[Piccard's 1st approximation]}$$
$$y_2(x) = y(x_o) + \int_{x_o}^{x} f(x, y_1(x)) dx \quad \text{[Picard's 2nd approximation]}$$
$$y_n(x) = y(x_o) + \int_{x_o}^{x} f(x, y_{n-1}(x)) dx \quad \text{[Picard's } n\text{th approximation]}$$

If the sequence $\{y_n\}_{n=1}^{\infty} \to y(x)$, then $y(x)$ is the unique solution to the IVP, and $|y_n(x) - y(x_o)| \le M|x - x_o|$

For an $\epsilon > 0$, we stop the iteration when

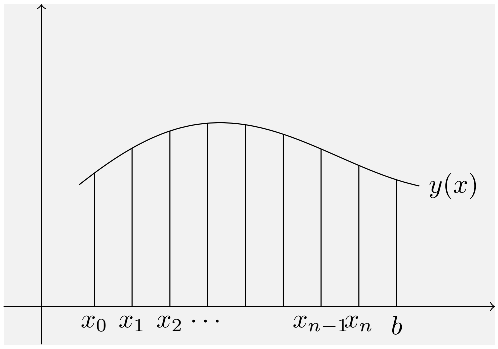
$$|y_n(x) - y_{n-1}(x)| < \epsilon, \quad \forall x \in [x_o, a].$$

2.3.2 Numerical methods for solving IVP

1. Single-step methods: Euler's method, Taylor's method, Runga-Kutta method
2. Multi-step methods

3 July 31, 2025 (Class 3)

3.1 Single step methods



We have

$$
\begin{aligned}
y' &= f(x,y) \\
y(x_o) &= y_o \\
x &\in [x_o, b] \\
N &:= \frac{b-x_o}{h} \\
S &:= \underbrace{\{x_0 < x_1 < x_2 < \ldots < x_N\}}_{\text{Grid points}} \text{ where } x_N = b
\end{aligned}
$$

,where h is the step size, N is the number of steps, and S is the set of grid points, and $f(x,y)$ satisfies the conditions of Picard's theorem.

The grid points are equally spaced, i.e., $h = x_{n+1} - x_n$ is constant for all n, where h is the step size.

By definition, A single step method is a method that computes the next value y_{n+1} using only the current value y_n and the current point x_n.

Single step methods are categorized in two types:

1. Explicit methods: The next value y_{n+1} is computed directly from the current value y_n and the function $f(x_n, y_n)$, i.e., $y_{n+1} = y_n + h\phi(x_n, y_n)$.

2. Implicit methods: The next value y_{n+1} is computed by solving an equation that involves both y_{n+1} and y_n, i.e., $y_{n+1} = y_n + h\phi(x_n, y_n, x_{n+1}, y_{n+1}; h)$.

where ϕ is a continuous function of its arguments, and assumed to satisfy Lipschitz condition w.r.t. y, (also called as the increment function).

3.2 Taylor's method

$$y' = f(x,y), \quad y(x_o) = y_o, x \in [x_o, b]$$

Assumptions:

1. $f(x,y)$ satisfies the conditions of Picard's theorem.

2. The solution $y(x)$ is sufficiently smooth, i.e., it has continuous derivatives of order $(p+1)$ on the interval $[x_o, b], p \geq 1$.

Then the solution $y(x)$ can be approximated by the Taylor series expansion around the point x_o:

$$y(x) = y(x_o) + (x - x_o)y'(x_o) + \frac{(x - x_o)^2}{2!}y''(x_o) + \ldots + \frac{(x - x_o)^p}{p!}y^{(p)}(x_o) + \mathcal{O}((x - x_o)^{p+1})$$

The remainder term $\mathcal{O}((x - x_o)^{p+1})$ represents the error in the approximation, which is of order $(p+1)$, and the specific form is,

$$\frac{(x - x_o)^{p+1}}{(p+1)!}y^{(p+1)}(\xi), \quad \xi \in (x_o, x)$$

Derivation of remainder term. To get the specific form of the remainder term, we can use the integral form of the remainder in Taylor's theorem:

$$\begin{aligned} \frac{1}{p!}\int_{x_o}^{x}(x-t)^p y^{(p+1)}(t)dt &= \frac{1}{p!}y^{(p+1)}(\xi)\int_{x_o}^{x}(x-t)^p dt \quad \text{[by Integral Mean Value Theorem]} \\ &= \frac{1}{p!}y^{(p+1)}(\xi)\left[\frac{(x-x_o)^{p+1}}{(p+1)}\right] \\ &= \frac{(x-x_o)^{p+1}}{(p+1)!}y^{(p+1)}(\xi), \quad \xi \in (x_o, x) \end{aligned}$$

□

3.2.1 Truncation error

We have the truncation error as

$$\mathcal{E}_n = \frac{h^{p+1}}{(p+1)!}y^{(p+1)}(\xi_n)$$

Derivation of truncation error. At $x = x_{n+1}$,

$$y(x_{n+1}) = y(x_n) + (x_{n+1} - x_n)y'(x_n) + \ldots + \frac{(x_{n+1} - x_n)^p}{p!}y^{(p)}(x_n) + \frac{(x_{n+1} - x_n)^{p+1}}{(p+1)!}y^{(p+1)}(\xi_n).$$

Letting $h = x_{n+1} - x_n$, for $n = 0, 1, \ldots, N-1$, we can rewrite the above equation as:

$$\begin{aligned} y(x_{n+1}) &= y(x_n) + hy'(x_n) + \frac{h^2}{2!}y''(x_n) + \ldots + \frac{h^p}{p!}y^{(p)}(x_n) + \frac{h^{p+1}}{(p+1)!}y^{(p+1)}(\xi_n) \\ &= y(x_n) + hy'(x_n) + \frac{h^2}{2!}y''(x_n) + \ldots + \frac{h^p}{p!}y^{(p)}(x_n) + \mathcal{E}_n \quad n = 0, 1, \ldots, N-1, \end{aligned}$$

where $\mathcal{E}_n = \frac{h^{p+1}}{(p+1)!}y^{(p+1)}(\xi_n)$ is the **local** truncation error at the nth step, with order $\mathcal{O}(h^{p+1})$. □

3.2.2 General form and approximation

We can express the relationship for the exact solution in a form that resembles our numerical method,

$$y_{n+1} = y_n + h\left(y'(x_n) + \frac{h}{2!}y''(x_n) + \ldots + \frac{h^{p-1}}{(p)!}y^{(p)}(x_n)\right) + \mathcal{E}_n, \quad n = 0, 1, \ldots, N-1.$$

By dropping the local error term $\mathcal{E}_n$, we get the Taylor's method for solving the IVP, with order p,

$$y_{n+1} = y_n + h\phi(x_n, y_n; h), \quad n = 0, 1, \ldots, N-1.$$

where

$$\phi(x_n, y_n; h) = y'(x_n) + \frac{h}{2!}y''(x_n) + \ldots + \frac{h^{p-1}}{(p)!}y^{(p)}(x_n)$$

is the increment function, which is a polynomial of degree $p-1$ in h.

3.3 Computation of higher order derivatives of $y(x)$ at $x = x_n$

Assumption: The function $f(x, y)$ is sufficiently smooth.

$$\begin{aligned} y'(x_n) &\approx y'_n = f(x_n, y_n) \\ y'' &\approx \frac{df}{dx} = \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y}y' = \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y}f(x_n, y_n) \end{aligned}$$

TODO: complete the computation of higher order derivatives

4 August 4, 2025 (Class 4)

TODO: complete this section

4.1 Truncation error

There are two types of truncation errors:

1. Local truncation error
2. Global truncation error

4.1.1 Local truncation error

We have T_n as the local truncation error at the nth step, and

$$\epsilon_n = hT_n = \frac{h^{p+1}}{(p+1)!}y^{(p+1)}(x_n + \theta h), \quad \text{where } 0 < \theta < 1,$$

where

$$T_n = \frac{h^p}{(p+1)!}y^{(p+1)}(x_n + \theta h).$$

From the Taylor's series expansion we have,

$$\begin{aligned} y(x_{n+1}) &= y(x_n) + h\phi(x_n, y(x_n); h) + \epsilon_n \\ &= y(x_n) + h\phi(x_n, y(x_n); h) + hT_n \end{aligned}$$

$$T_n = \frac{y(x_{n+1}) - y(x_n)}{h} - \phi(x_n, y(x_n); h)$$

4.1.2 Global truncation error

$$T = \max_{0 \le x \le N-1} |T_n|$$

If $\epsilon > 0$ is preassigned number to bound the truncation error, then

$$|hT_n| < \epsilon$$

$$\left| \frac{h^{p+1}}{(p+1)!} f^{(p+1)}(x_n + \theta h, y(x_n + \theta h)) \right| \le \epsilon$$

- For a given h and ϵ, the above equation will give p, i.e. the number of terms in the Taylor series method.
- If p and ϵ are specified, then it will give the upper bound on h.

4.1.3 Bound on truncation error

Given the local truncation error T_n for the Taylor's method, we can express it as follows:

$$\begin{aligned} T_n &= \frac{h^p}{(p+1)!} y^{(p+1)}(\xi_n) \\ |T_n| &\le \frac{h^p}{(p+1)!} M_{p+1}, \quad \text{where } M_{p+1} = \max_{x_0 \le \xi_n \le b} |y^{(p+1)}(\xi_n)| \end{aligned}$$

4.2 Global error

We define the global error as

$$e_n = y(x_n) - y_n,$$

where $y(x_n)$ is the exact value and y_n is the approximated value.

4.3 Rounding error

5 August 6, 2025 (Class 5)

5.1 Bound on global error in terms of truncation error

Theorem 5.1 (Global Error Bound for Single-Step Methods.): *Let the initial value problem (IVP) be defined on the interval $[a, b]$ by*

$$y'(x) = f(x, y(x)), \quad y(a) = y_0$$

and let $y(x)$ denote its unique, exact solution.

Consider a numerical approximation to this IVP generated by a general single-step method,

$$y_{n+1} = y_n + h\phi(x_n, y_n; h), \quad n = 0, 1, \ldots, N-1,$$

where $h = (b-a)/N$ is the uniform step size, $x_n = a+nh$ and y_n is the approximation to $y(x_n)$. We assume ϕ is a continuous function of its arguments, and satisfies Lipschitz condition (see Lipschitz continuity) with respect to its second argument, with Lipschitz constant L.

Then we have

$$|e_n| \leq \frac{T}{L}\left(e^{L(x_n - x_o)} - 1\right), \quad n = 0, 1, \ldots, N-1$$

where $T = \max\limits_{0 \leq n \leq N-1} |T_n|$ and T_n are the local truncation errors.

This result establishes that if the local truncation error T_n is small, the global error will also be small, with its growth controlled exponentially by the Lipschitz constant L and the width of the integration interval.

Proof of the bound on global error. We have,

$$\begin{aligned} T_n &= \frac{y(x_{n+1}) - y(x_n)}{h} - \phi(x_n, y(x_n); h) \\ \implies y(x_{n+1}) &= y(x_n) + h\phi(x_n, y(x_n); h) + hT_n \end{aligned}$$

From the single step method, we have,

$$y_{n+1} = y_n + h\phi(x_n, y_n; h)$$

Subtracting the two equations, we get,

$$\begin{aligned} (y(x_{n+1}) - y_{n+1}) &= (y(x_n) - y_n) + h\left(\phi(x_n, y(x_n); h) - \phi(x_n, y_n; h)\right) + hT_n \\ e_{n+1} &= e_n + h\left(\phi(x_n, y(x_n); h) - \phi(x_n, y_n; h)\right) + hT_n \\ |e_{n+1}| &\leq |e_n| + h\left|\phi(x_n, y(x_n); h) - \phi(x_n, y_n; h)\right| + |hT_n| \\ &\leq |e_n| + L_\phi h|y(x_n) - y_n| + |hT_n| \qquad \text{[Lipschitz continuity of } \phi] \\ &\leq |e_n| + L_\phi h|e_n| + hT \qquad \text{where } |T| = \max_{0 \leq n \leq N-1} |T_n| \\ |e_{n+1}| &\leq |e_n|(1 + L_\phi h) + hT, \quad n = 0, 1, \ldots, N-1 \end{aligned}$$

For $n = 0$,

$$\begin{aligned} |e_1| &\leq |e_0|(1 + hL_\phi) + hT \\ e_0 &= y(x_0) - y_0 = 0 \implies |e_1| \leq hT \end{aligned}$$

For $n = 1$,

$$\begin{aligned} |e_2| &\leq |e_1|(1 + hL_\phi) + hT \\ &\leq hT(1 + hL_\phi) + hT \\ &= 2hT + h^2 L_\phi T \\ &= \frac{T}{L_\phi}\left(2L_\phi h + h^2 L_\phi^2\right) \\ \implies |e_2| &\leq \frac{T}{L_\phi}\left((1 + hL_\phi)^2 - 1\right) \end{aligned}$$

For general n,

$$
\begin{aligned}
|e_n| &\le \frac{T}{L_\phi}\left((1+hL_\phi)^n - 1\right) \quad n = 0, 1, \ldots, N-1 \\
&\le \frac{T}{L_\phi}\left(e^{L_\phi(x_n - x_o)} - 1\right) \quad [\text{since } (1+x)^n \le e^{nx} \text{ for } x > -1, n > 0]
\end{aligned}
$$

□

5.2 Consistency of a single step method

The general single step method is said to be consistent with the given IVP if

TODO: complete this section

5.3 Order of a single step method

TODO: complete this section

6 August 7, 2025 (Class 6)

6.1 Convergence of a single step method

Assume that the IVP satisfies the conditions of Picard's theorem, and also that its approximation generated from the single step method when $h < h_0$ lies in the region $\mathcal{D}$. Assume further that the function ϕ is Lipschitz continuous in y on $\mathcal{D} \times [0, h_0]$ and satisfies the consistency condition $\phi(x, y; 0) = f(x, y)$. Then for y_n generated by the single step method, we have, for any fixed point $x \in [x_0, b]$,

$$\lim_{h \to 0} y_n = y(x).$$

6.1.1 Proof of convergence of a single step method

We define

$$h := \frac{b - x_0}{N}, \quad N \in \mathbb{N}^+$$

and N is sufficiently large such that $h < h_0$.

TODO: write the proof

7 August 11, 2025 (Class 7)

7.1 Euler method

From the first order taylor's method method we can get the Euler method.

$$y_{n+1} = y_n + hy'_n = y_n + hf(x_n, y_n)$$

Assumption: The step size h is taken to be very small to bound the error of $O(h^2)$ in the taylor series method

$$\begin{aligned} y(x_{n+1}) &= y(x_n) + hy'(x_n) + O(h^2) \\ &= y(x_n) + hf(x_n, y(x_n)) + \frac{h^2}{2!}y''(\xi), \quad x_n < \xi < x_{n+1} \end{aligned}$$

7.1.1 Truncation error of Euler method

$$\begin{aligned} T_n &= \frac{y(x_{n+1}) - y(x_n)}{h} - \phi(x_n, y(x_n); h) \\ &= \frac{1}{h}\left(y(x_n) + hy'(x_n) + \frac{h^2}{2!}y''(\xi) - y(x_n)\right) - f(x_n, y(x_n)) \\ &= \frac{h}{2!}y''(\xi) \qquad\qquad \because y'(x_n) = f(x_n, y(x_n)) \text{ from IVP} \end{aligned}$$

Therefore,

$$|T_n| \le Kh, \quad \text{where } K = \frac{1}{2}\max_{\xi\in[x_0,b]} |y''(\xi)|.$$

Euler method is of order 1.

7.1.2 Global error of Euler method

TODO: complete this section

8 August 13, 2025 (Class 8)

The differential equation is given by

$$\frac{dy}{dx} = f(x, y)$$

From this, we have

$$\begin{aligned} \int_{x_n}^{x_{n+1}} y'dx &= \int_{x_n}^{x_{n+1}} f(x, y)dx \\ y(x_{n+1}) &= y(x_n) + \int_{x_n}^{x_{n+1}} f(x, y)dx \\ &= y(x_n) + f(\xi_n, y(\xi_n))\int_{x_n}^{x_{n+1}} dx, \quad \text{where } x_n < \xi_n < x_{n+1} \\ &= y(x_n) + (x_{n+1} - x_n)f(\xi_n, y(\xi_n)) \\ &= y(x_n) + hf(\xi_n, y(\xi_n)) \end{aligned}$$

$$y(x_{n+1}) = y(x_n) + hf(x_n + \theta h, y(x_n + \theta h)), \quad \text{where } 0 < \theta < 1$$

8.1 Euler method, $\theta = 1$

For $\theta = 0$, we have

$$y_{n+1} = y_n + hf(x_n, y_n).$$

8.2 Backward Euler method, $\theta = 1$

For $\theta = 1$, we have

$$y_{n+1} = y_n + hf(x_n + h, y(x_n + h)).$$

Explicit form,

$$y_{n+1} = y_n + hf(x_n, y_n + hf(x_n, y_n)).$$

8.3 Modified Euler method, $\theta = 1/2$

$$y_{n+1} = y_n + hf(x_n + h/2, y(x_n + h/2))$$

$$y_{n+1} = y_n + hf\left(x_n + \frac{h}{2}, y_n + \frac{h}{2}f(x_n, y_n)\right)$$

8.4 Trapezium method

$$y_{n+1} = y_n + \frac{h}{2}\left(f(x_n, y_n) + f(x_{n+1}, y_{n+1})\right)$$

2nd order, implicit method

8.5 Heun's method / Euler Cauchy method

$$y_{n+1} = y_n + \frac{h}{2}\left(f(x_n, y_n) + f(x_n + h, y_n + hf(x_n, y_n))\right)$$

Explicit method

8.6 Runge-Kutta methods

9 August 14, 2025 (Class 9)

10 August 18, 2025 (Class 10)

11 August 20, 2025 (Class 11)

12 August 21, 2025 (Class 12)

13 August 25, 2025 (Class 13)

14 August 28, 2025 (Class 14)

15 September 1, 2025 (Class 15)

16 I will divide the notes into classes later.....

16.1 Root finding

16.1.1 Types of roots

simple multiple.

16.1.2 Methods of root finding

1. Direct methods
2. Iterative methods

16.2 Iterative methods of root finding

A sequence of iterative methods is said to converge to the root ξ of the function $f(x)$ if

$$\lim_{k\to\infty} |x_k - \xi| = 0$$

16.3 Initial approximation

Two methods used for initial approximation are

1. Graphical approximation
2. Using intermediate value theorem

16.4 Some theorems and definitions

Theorem 16.1: *Let f be a real valued function defined and continuous on a bounded closed interval $[a, b]$ of the real line. Assume, further that $f(a)f(b) \leq 0$; then there exists $\xi \in [a, b]$ such that $f(\xi) = 0$.*

Theorem 16.2 (Intermediate value theorem): *Suppose that f is a real-valued function, defined and continuous on the closed interval $[a, b] \in \mathbb{R}$. Then f is bounded function on the interval $[a, b]$ and if y is any number such that*

$$\inf_{x\in[a,b]} f(x) \leq y \leq \sup_{x\in[a,b]} f(x),$$

then there is a number $\xi \in [a, b]$ such tht $f(\xi) = y$.

Theorem 16.3 (Brouwer fixed point theorem): *TODO: write this*

Definition 16.4 (Fixed point iteration): *Suppose that g is a real valued function, defined and continuous on a bounder closed interval $[a, b]$ of the real line, and assume that $g(x) \in [a, b] \forall x \in [a, b]$. Given that $x_0 \in [a, b]$, the recursion defined by*

$$x_{k+1} = g(x_k), k = 0, 1, 2, \ldots$$

is called a simple iteration; the number x_k are referred to as iterates.

If the sequence $\{x_k\}$ defined above converges, the limit must be a fixed point of the function g, since g is continuous on a closed interval, i.e.

$$\lim_{k\to\infty} x_k = \xi \implies \xi = g(\xi).$$

Theorem 16.5 (A Priori Iteration Bound for Contractions): *TODO: write this simple iteration $x_{k+1} = g(x_k)$, contraction map on $[a, b]$, $x_0 \in [a, b]$, tolerance $\varepsilon > 0$,*

$$k_o(\varepsilon) := \inf_{k\in\mathbb{I}_+} \{k \text{ s.t. } |x_k - \xi| \leq \varepsilon \forall k \geq k_o(\varepsilon)\}.$$

Then,

$$k_o(\varepsilon) \leq \left(\frac{\ln|x_1 - x_0| - \ln(\varepsilon(1 - L))}{\ln(1/L)}\right) + 1,$$

where $L \in [0, 1)$, is the Lipschitz constant of the contraction mapping g.

Theorem 16.6 (Banach fixed point theorem): *TODO: write this*

Proof. TODO: write this ∎

17 September 3, 2025 (Class 16)

18 September 4, 2025 (Class 17)

19 September 8, 2025 (Class 18)

20 I will divide the notes into classes later.....

20.1 Stable and unstable fixed points

Definition 20.1 (Stable and unstable fixed points): *Let $g : \mathbb{R} \to \mathbb{R}$ be continuous, and let $\xi \in \mathbb{R}$ be a fixed point of g (i.e. $g(\xi) = \xi$). We say ξ is a* stable fixed point *(or* attracting fixed point*) if there exists a neighborhood U of ξ such that for all $x_0 \in U$, the iterates defined by*

$$x_{k+1} = g(x_k), \quad k \geq 0,$$

converge to ξ. Conversely, if for every neighborhood U of ξ there exists some $x_0 \in U$ such that the sequence $\{x_k\}$ does not converge to ξ, then ξ is called an unstable fixed point.

In the differentiable case, a fixed point ξ is stable if $|g'(\xi)| < 1$, unstable if $|g'(\xi)| > 1$, and inconclusive if $|g'(\xi)| = 1$.

20.2 Bisection method

20.3 Relaxation base iterative method

Let $f : I \to \mathbb{R}$ be a function with a simple root ξ in an open interval $I \subseteq \mathbb{R}$. The goal is to find ξ such that $f(\xi) = 0$. The relaxation-based iterative method generates a sequence $\{x_k\}$ from an initial guess x_0 using the formula:

$$x_{k+1} = x_k - \lambda f(x_k), \quad k \geq 0$$

where $\lambda \neq 0$ is a constant parameter called the **relaxation parameter**.

This method can be viewed as a **fixed-point iteration**, $x_{k+1} = g(x_k)$, with the iteration function defined as:

$$g(x) = x - \lambda f(x)$$

A point ξ is a fixed point of g if and only if it is a root of f, since $g(\xi) = \xi \iff \xi - \lambda f(\xi) = \xi \iff f(\xi) = 0$.

20.3.1 Convergence of the iterates in relaxation base iterative method

Theorem 20.2 (Local Convergence of the Relaxation Method): *Let f be a continuously differentiable function in a neighborhood of a simple root ξ (i.e., $f(\xi) = 0$ and $f'(\xi) \neq 0$). The sequence defined by $x_{k+1} = x_k - \lambda f(x_k)$ converges to ξ for any initial guess x_0 sufficiently close to ξ, provided the relaxation parameter λ satisfies the condition:*

$$0 < \lambda f'(\xi) < 2$$

Proof. The iteration converges locally to ξ if the iteration function $g(x) = x - \lambda f(x)$ is a contraction mapping in a neighborhood of ξ. A sufficient condition for this is that g is continuously differentiable and $|g'(\xi)| < 1$.

Differentiating,

$$g'(x) = \frac{d}{dx}(x - \lambda f(x)) = 1 - \lambda f'(x)$$

Applying the convergence condition at the fixed point ξ,

$$|g'(\xi)| < 1 \implies |1 - \lambda f'(\xi)| < 1$$

This is equivalent to the pair of inequalities:

$$\begin{aligned} -1 < 1 - \lambda f'(\xi) < 1 \\ -2 < -\lambda f'(\xi) < 0 \\ 0 < \lambda f'(\xi) < 2 \end{aligned}$$

This proves the theorem. The strict inequalities are crucial; if $|g'(\xi)| = 1$, convergence is not guaranteed.

Conclusion The condition $0 < \lambda f'(\xi) < 2$ implies that λ must have the **same sign** as $f'(\xi)$. The convergence is fastest when $g'(\xi)$ is closest to zero, which occurs when $1 - \lambda f'(\xi) = 0$, or $\lambda = 1/f'(\xi)$. This observation forms the basis for Newton's method. ☐

20.4 Newton's method

20.5 Secant method

21 Mid term

22 October 6, 2025 (Class __)

Lagrange interpolating polynomial $p_x(x)$

22.1 Error term in Lagrange interpolation

Theorem 22.1 (Error in Lagrange interpolation): *Let $k \in \mathbb{N}$ and $f : \mathbb{R} \to \mathbb{R}$. Furthermore, $f \in \mathbb{C}([a, b])$ and $f^{(k+1)} \in \mathbb{C}([a, b])$. Then, if p_x is the Lagrange interpolation of f of degree at most k, we have,*

$$R(x) = f(x) - p_x(x) = \frac{f^{(k+1)}(\xi)}{(k+1)!} \prod_{i=0}^{k} (x - x_i), \quad x_0 < \xi < x_k$$

and

$$|R(x)| \leq \frac{1}{(k+1)!} \max_{\xi \in [x_0, x_k]} |f^{(k+1)}(\xi)| \max_{x \in [x_0, x_k]} \left| \prod_{i=0}^{k} (x - x_i) \right|.$$

We obtain a looser but simpler bound taking $\max_{x\in[x_0,x_k]}\left|\prod_{i=0}^{k}(x-x_i)\right| \leq (x_k-x_0)^{k+1}$,

$$|R(x)| \leq \frac{(x_k-x_0)^{k+1}}{(k+1)!} \max_{\xi\in[x_0,x_k]} |f^{(k+1)}(\xi)|.$$

Proof. TODO □

23 October 8, 2025 (Class __)

The previous theorem gives high errors in some cases. Weierstrass approximation theorem.

23.1 Hermite interpolation

Theorem 23.1 (Hermite interpolation): *For an unknown function $f:\mathbb{R}\to\mathbb{R}$, we have for data points $\{x_i\}_0^n$, $\{y_i=f(x_i)\}_0^n$ and $\{z_i=f'(x_i)\}_0^n$. Then we have the Hermite interpolation polynomial p of degree $2n+1$,*

$$p_{2n+1}(x) = \sum_{k=0}^{n} (H_k(x)y_k + K_k(x)z_k)$$

where

$$H_k(x) = (l_k(x))^2\,(1-2l_k'(x_k)(x-x_k))$$
$$K_k(x) = (l_k(x))^2\,(x-x_k)$$

24 October 10, 2025, (Class __)

24.1 Solution of systems of linear equations

24.2 Forward & Backward substitution

24.3 Cramer's rule

For a linear system of equations

$$Ax = b,$$

given $\det(A)\neq 0$, we have

$$x_i = \frac{\det(A_i)}{\det(A)}, \quad i=1,2,\ldots,n$$

We define A_i as

$$A_i = \begin{bmatrix} a_{11} & \cdots & b_1 & \cdots & a_{1n} \\ a_{21} & \cdots & b_2 & \cdots & a_{2n} \\ \vdots & & \vdots & & \vdots \\ a_{n1} & \cdots & b_n & \cdots & a_{nn} \end{bmatrix}$$

where b replaces the i-th column.

24.4 Gauss' elimination

24.5 Triangular factorization methods

24.5.1 LU decomposition based method

Using `[LU Decomposition]`, we have

$$A = LU.$$

We can decompose the original system of linear equations in two,

$$\begin{aligned} Ux &= z \\ Lz &= b \end{aligned}$$

The solution to the original system can be obtained by first solving $Lz = b$ using **forward substitution** (since L is lower triangular), and then solving $Ux = z$ using **backward substitution** (since U is upper triangular).

Remark 24.1: *This method can fail for two primary reasons. First, the factorization $A = LU$ (without pivoting) may not exist if a leading principal minor of A is singular. Second, the substitution steps will fail if any diagonal element L_{ii} or U_{ii} is 0. This computationally confirms that A is singular, since* $\det(A) = \det(L)\det(U)$.

24.5.2 Cholesky decomposition based method

Using `[Cholesky Decomposition]`, we have

$$A = LL^T.$$

We can decompose the original system of linear equations in two,

$$\begin{aligned} L^T x &= z \\ Lz &= b \end{aligned}$$

The solution to the original system can be obtained by first solving $Lz = b$ using **forward substitution** (since L is lower triangular), and then solving $L^T x = z$ using **backward substitution** (since L^T is upper triangular).

Remark 24.2: *This method fails if the matrix A is not symmetric and positive definite. Computationally, this failure occurs during the factorization (not the substitution) if the algorithm requires taking the square root of a non-positive number to compute a diagonal element L_{ii}.*

24.6 Error analysis for direct methods

24.7 See also

24.8 References